# Marco Notaro

*Curriculum Vitae*

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## Education

Oct 2015-Feb 2019 **Ph.D. in Computer Science**, *Thesis Title: "Hierarchical Ensemble Methods for Ontology-based Predictions in Computational Biology"*, Milan University, Computer Science Department, Advisor: Prof. Giorgio Valentini, Co-Advisor: Dr. Matteo Re.

Oct 2012-Feb 2015 **M.Sc. in Molecular Biotechnology and Bioinformatics (Curriculum Bioinformatics)**, *Thesis Title: "Hierarchical Ensemble Methods for Biological Ontologies Structured as Directed Acyclic Graphs"*, Milan University, Computer Science Department, Advisor: Prof. Giorgio Valentini, Co-Advisor: Dr. Matteo Re – Final Grade: 110/110.

Oct 2009-Oct 2012 **B.Sc. in Industrial and Environmental Biotechnology (Curriculum Bioinformatics)**, *Thesis Title: "High Resolution Maps of Methylation and Acetylation of Histones in the Human Genome"*, Milan University, Bioscience Department, Advisor: Prof. Giulio Pavesi – Final Grade: 101/110.

Sept 2004-July 2009 **Scientific High School Diploma**, Liceo Scientifico Galileo Galilei, Caravaggio (Bergamo), IT.

## Research and Work Experiences

Feb 2019-Present **Collaborator**, in the project "Data Model of Next-Generation DataBase Management System (DBMS)", Milan University, Computer Science Department. Supervisor: Prof. Marco Mesiti.

Oct 2015-Feb 2019 **PhD Research Activity**, at Computational Biology and Bioinformatics Group, Milan University, Computer Science Department, under the supervision of Prof. Giorgio Valentini. Some relevant research experiences alongside my PhD: • I was co-advisor of a master thesis in Computer Science; • I did Bioinformatics lectures for M.Sc. courses in Computer Science (link).

Mar 2018-May 2018 **Visiting PhD Student**, at CITIC-UGR, Centro de Investigación en Tecnologías de la Información y de las Cominucaciones, Granada University, Spain, under the supervision of Dr. Carlos Cano Gutierrez.

Sept 2016-Jan 2017 **CAFA3 Challenge**, international challenge for the prediction of protein function and for the prediction of genes associated with abnormal phenotypes.

June 2016-Aug 2016 **Visiting PhD Student**, at Charité - Universitätsmedizin Berlin, Institute for Human and Medical Genetics, joining in the *Computational Biology Group* under the supervision of Prof. Peter Robinson.

Oct 2015-Jan 2016 **Assistant**, for the course "Statistics and Data Analysis" (R Laboratory), Milan University, Computer Science Department, Advisor: Prof. Dario Malchiodi (link).

Mar 2015-May 2015 **Assistant**, for the course "Data Analysis" (R Laboratory), Milan University, Computer Science Department, Advisor: Anna Mari Zanaboni (link).

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| Feb 2014-Feb 2015 | **Master Degree Internship**, at Computational Biology and Bioinformatics Group, Milan University, Computer Science Department, Advisor: Prof. Giorgio Valentini, Co-Advisor: Dr. Matteo Re. |

## Summer Schools and International Meeting

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| 12-16 Sept 2016 | **Big Data for Precision Medicine and The 4th i2b2 European Meeting**, Pavia, IT. |

## Speaker at International Conferences

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| 7-9 Sept 2016 | **CIBB2017**, Cagliari, IT, Took part as speaker by presenting the following works 1. Ensembling Descendant Term Classifiers to Improve Gene - Abnormal Phenotype Predictions; 2. Disease Genes must Guide Data Source Integration in the Gene Prioritization Process. |

## Language Skills and Certifications

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| Italian | Mother Tongue. |
| English | Upper Intermediate Level, (Certified by Shenker Method -2013-). |

## Computational Skills

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| Main Programming Languages | • **R**: machine learning; statistics and data analysis; data manipulation and data visualization (ggplot2) • **Perl**: parsing and analysis of raw text files; pattern matching by regular expression • **Bash**: scheduling computational experiments and executing commands in Linux environment |
| Side Programming Languages | **JavaScript**: misc. and basic knowledge for maintaining my website • **Python**: misc. and basic knowledge for scientific usage (as numpy, scipy) • **MATLAB and C++**: some basic and misc. knowledge |
| Markup Languages | • **HTML** and **CSS**: for maintaining my website • **LaTex**: for writing scientific papers and reports |
| Software Libraries | • **HEMDAG** an R package available both on biconda and cran repository alongside a step-by-step tutorial (link); • **HPOparser/GOparser** (perl modules) (note: a more general perl module, named **OBOparser**, will be submitted to *cpan* shortly) For more details about what these packages do, please have a look at my Software Page |
| Operating systems | • Linux (Ubuntu); • Windows |

## Research Interests

My research interests touch the fields of:

- Bioinformatics
- Computational Biology
- Machine Learning

To know further details about my research interests, please get a glimpse of my Research Page.

## Awards

**Prediction of Human Phenotype Ontology terms by means of Hierarchical Ensemble Methods**: awarded by the International Medical Informatics Association (IMIA) as one of the five best "Knowledge Representation and Management" (KRM) papers of 2017 in the field of Medical Informatics (link)

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## Publications

Note: an updated list of all my papers can be also found on my Publications Page

## International Peer-Reviewed Journals

5. **Notaro M**, Frasca M, Petrini A, Valentini G. HEMDAG: a scalable and flexible state-of-the-art tool outperforming flat learning predictions; 2019. Manuscript in preparation.

4. Zhou N, Jiang Y, . . . , **Notaro M**, . . . , Radivojac P, Friedberg I. The CAFA challenge reports improved protein function prediction and new functional annotations for hundreds of genes through experimental screens. bioRxiv. 2019;Cold Spring Harbor Laboratory. doi:10.1101/653105. Available from: https://www.biorxiv.org/content/early/2019/05/29/653105.

3. Perlasca P, Frasca M, Ba CT, **Notaro M**, Petrini A, Grossi G, Gliozzo J, Valentini G, Mesiti M. UNIPred-Web: a Web Tool for the Integration and Visualization of Biomolecular Networks for Protein Function Prediction; 2019. Accepted for publication.

2. Frasca M, Grossi G, Gliozzo J, Mesiti M, **Notaro M**, Perlasca P, Petrini A, Valentini G. A GPU-based algorithm for fast node label learning in large and unbalanced biomolecular networks. BMC Bioinformatics. 2018;19(10):353. doi:10.1186/s12859-018-2301-4. Available from: https://doi.org/10.1186/s12859-018-2301-4.

1. **Notaro M**, Schubach M, Robinson PN, Valentini G. Prediction of Human Phenotype Ontology terms by means of hierarchical ensemble methods. BMC Bioinformatics. 2017;18(1):449. doi:10.1186/s12859-017-1854-y. Available from: https://doi.org/10.1186/s12859-017-1854-y.

## Proceedings of International Conferences and Peer-Reviewed Book Chapters

4. **Notaro M**, Schubach M, Frasca M, Mesiti M, Robinson PN, Valentini G. Ensembling Descendant Term Classifiers to Improve Gene - Abnormal Phenotype Predictions. In: Bartoletti M, Barla A, Bracciali A, Klau GW, Peterson L, Policriti A, Tagliaferri R, editors. Computational Intelligence Methods for Bioinformatics and Biostatistics. Cham: Springer International Publishing; 2019. doi:10.1007/978-3-030-14160-8_8. Available from: https://doi.org/10.1007/978-3-030-14160-8_8.

3. Frasca M, Fontaine JF, Valentini G, Mesiti M, **Notaro M**, Malchiodi D, Andrade-Navarro MA. Disease–Genes Must Guide Data Source Integration in the Gene Prioritization Process. In: Bartoletti M, Barla A, Bracciali A, Klau GW, Peterson L, Policriti A, Tagliaferri R, editors. Computational Intelligence Methods for Bioinformatics and Biostatistics. Cham: Springer International Publishing; 2019. doi:10.1007/978-3-030-14160-8_7. Available from: https://doi.org/10.1007/978-3-030-14160-8_7.

2. Robinson PN, Frasca M, Köhler S, **Notaro M**, Re M, Valentini G. A Hierarchical Ensemble Method for DAG-Structured Taxonomies. In: Schwenker F, Roli F, Kittler J, editors. Multiple Classifier Systems. Cham: Springer International Publishing; 2015. doi:10.1007/978-3-319-20248-8_2. Available from: https://doi.org/10.1007/978-3-319-20248-8_2.

1. Valentini G, Köhler S, Re M, **Notaro M**, Robinson PN. Prediction of Human Gene - Phenotype Associations by Exploiting the Hierarchical Structure of the Human Phenotype Ontology. In: Ortuño F, Rojas I, editors. Bioinformatics and Biomedical Engineering. Cham: Springer International Publishing; 2015. doi:10.1007/978-3-319-16483-0_7. Available from: https://doi.org/10.1007/978-3-319-16483-0_7.

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